

Software Requirements Specification (SRS)

AI Recipe Generator Web Application

1. Introduction

Purpose: Develop an AI-powered recipe generation platform that helps users create recipes using available ingredients, cuisine preferences, dietary restrictions, and meal types.

Scope: User authentication, AI recipe generation, recipe storage, search, filtering, user profiles, and deployment on Vercel/Render.

2. Product Overview

Frontend: React.js, Tailwind CSS, Axios, React Router.

Backend: Node.js, Express.js.

Database: MongoDB Atlas.

AI Service: GROQ API.

Version Control: GitHub.

Deployment: Vercel (Frontend), Render (Backend).

3. Stakeholders and Users

Guest Users: Generate limited recipes.

Registered Users: Save and manage recipes.

Administrators: Monitor users, API usage, and system analytics.

4. Functional Requirements

FR-1 Registration

FR-2 Login

FR-3 Logout

FR-4 Generate Recipe

FR-5 Regenerate Recipe

FR-6 Save Recipe

FR-7 View Saved Recipes

FR-8 Delete Recipe

FR-9 Search Recipes

FR-10 Filter Recipes

FR-11 Update Profile

FR-12 Change Password

5. Detailed Use Cases

Generate Recipe: User enters ingredients and preferences. Backend sends prompt to GROQ API. AI returns recipe title, ingredients, instructions, cooking time, and optional nutrition details.

Save Recipe: Authenticated users save generated recipes in MongoDB.

6. Non-Functional Requirements

Performance: Page load <3 seconds, AI response <10 seconds.

Security: JWT, bcrypt, HTTPS, environment variables.

Availability: 99% uptime.

Scalability: Support 1000+ concurrent users.

7. System Architecture

User -> React Frontend -> Express API -> GROQ API / MongoDB Atlas -> Response to User.

8. Database Design

Users Collection: id, name, email, password, avatar, createdAt.

Recipes Collection: id, userId, title, ingredients, instructions, cuisine, mealType, dietType, createdAt.

9. API Specifications

POST /api/auth/register

POST /api/auth/login

GET /api/auth/profile

POST /api/recipe/generate

POST /api/recipe/save

GET /api/recipe/history

DELETE /api/recipe/:id

10. User Interface Requirements

Responsive design for desktop, tablet, and mobile.

Pages: Landing Page, Login, Register, Dashboard, Recipe Generator, Saved Recipes, Profile.

11. Deployment Requirements

Frontend deployment through Vercel connected to GitHub.

Backend deployment through Render.

MongoDB Atlas cloud database.

Environment variables for API keys and secrets.

12. Testing Strategy

Unit Testing, API Testing, Integration Testing, UI Testing, User Acceptance Testing.

13. Future Enhancements

AI-generated recipe images, multilingual support, voice input, grocery list generation, nutrition analysis, meal planning, social sharing.

14. Success Criteria

95% successful recipe generation rate, secure authentication, responsive UI, successful deployment and production readiness.